Class	Adm No

Candidate Name: ______________________________

H

millennia institute

2021 Prelim Examination
Pre-university 3

H2 Biology **9744/03**

Paper 3 Long Structured and Free-response Questions **8 September 2021**

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer all questions in the space provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**/25**
2	**/10**
3	**/15**
Section B	**/25**
Total	**/75**

This question paper consists of 22 printed pages, including 1 blank page.

Section A

Answer **all** questions in this section.

1. Adipose tissue, which is composed of cells known as adipocytes, stores large quantities of triglycerides and functions as an energy storage tissue. Adipocytes synthesise triglyceride lipase (ATGL), an enzyme that catalyses the formation or breakdown of triglycerides.

(a)

(i) State the bond that is broken when triglyceride is hydrolysed.

.... **[1]**

Linoleic acid is an unsaturated fatty acid that is found in some triglycerides and some phospholipids.

Fig. 1.1 shows a molecule of linoleic acid.

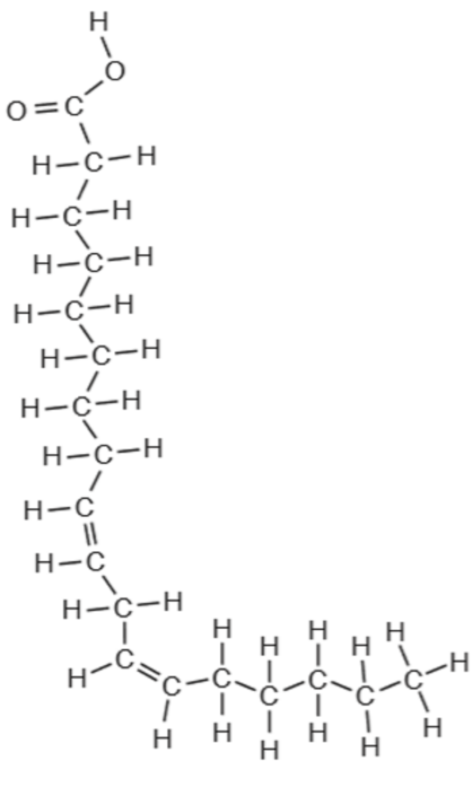

Fig. 1.1

The composition of cell membranes of plants changes in response to changes in temperature.

At the start of the cold season, there is change in the proportion of phospholipids with linoleic acid in the chickpea, *Cicer arietinum*. Chickpea plants that do not make this change do not survive.

(ii) Describe how the proportion of phospholipids with linoleic acid changes and explain how this helps plants, such as chickpea, survive the cold season.

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.... **[3]**

(b) The balance between triglyceride formation and breakdown is controlled by hormones. Fig. 1.2 is a summary of events occurring in an adipocyte when glycogen energy stores have been used up.

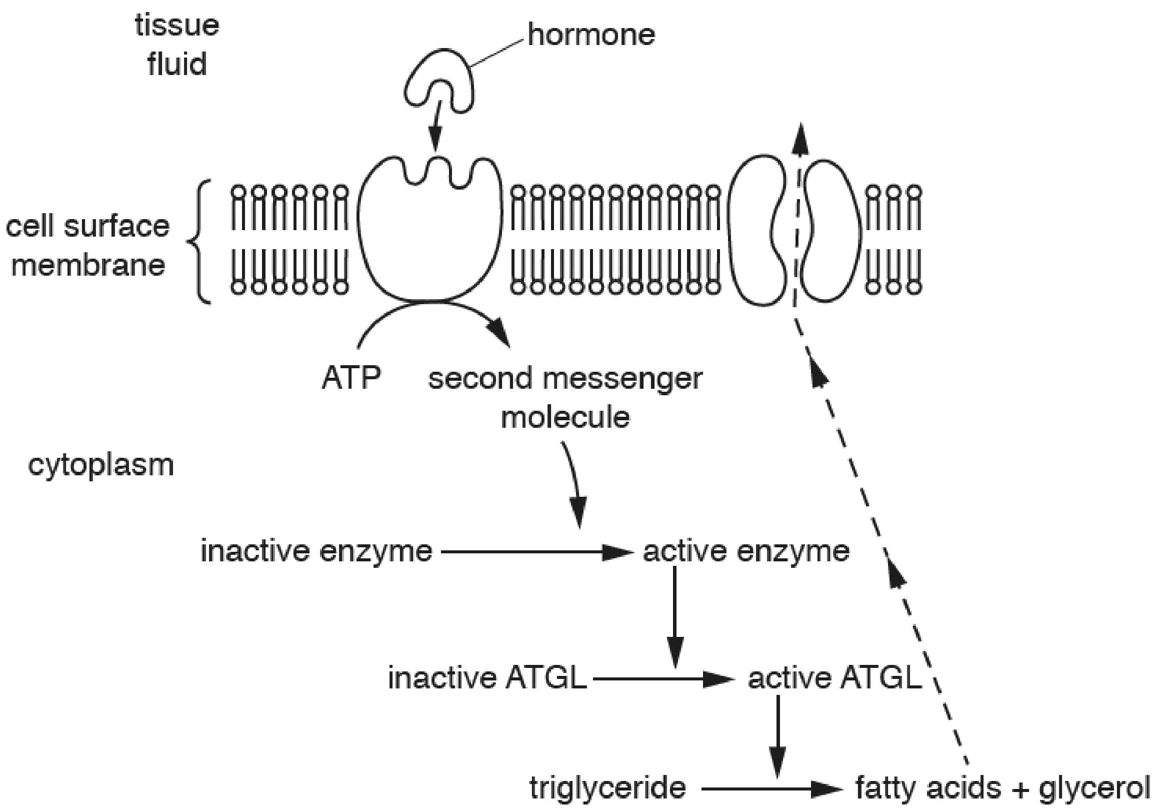

Fig. 1.2

The events shown in Fig. 1.2 is an example of cell signalling within the body.

(i) With reference to Fig. 1.2, outline how the hormone control the balance between metabolism of triglyceride and glycogen reserves and suggest the importance of this for survival.

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Fatty acids leave the cells upon triglyceride hydrolysis.

(ii) Explain why the fatty acids need to exit via the process shown in Fig. 1.2.

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(c) The fatty acids released from adipocytes are transported in blood plasma and are taken up by cells. Although most cell types can metabolise fatty acid to synthesise ATP in the presence of oxygen, red blood cells cannot do this.

Suggest two reasons to why red blood cells cannot metabolise fatty acids to synthesise ATP.

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....**[2]**

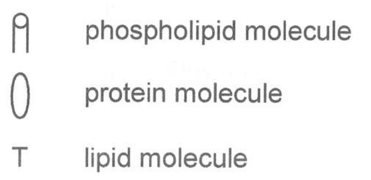
(d) Female mammals produce milk to feed newborn offspring.

(i) Milk contains a few substances, including lipids. These lipids, known as milk lipids, are synthesised in the lumen of the smooth endoplasmic reticulum within the milk-secreting cells.

These milk lipids collect in between the two layers of phospholipids that make up the phospholipid bilayer of the smooth endoplasmic reticulum membrane. During formation of droplet, the outer phospholipid layer of the bilayer pinches off. The droplet consists of a single layer of phospholipids with some membrane proteins and contain milk lipids.

Sketch the arrangement of molecules in one complete droplet.

You should use the following symbols.



[3]

The droplet moves through the cytoplasm to reach the surface of the cell, where it is secreted as a milk fat globule into a cavity within a mammary gland.

Fig. 1.3 shows the stages in this process.

Not to scale

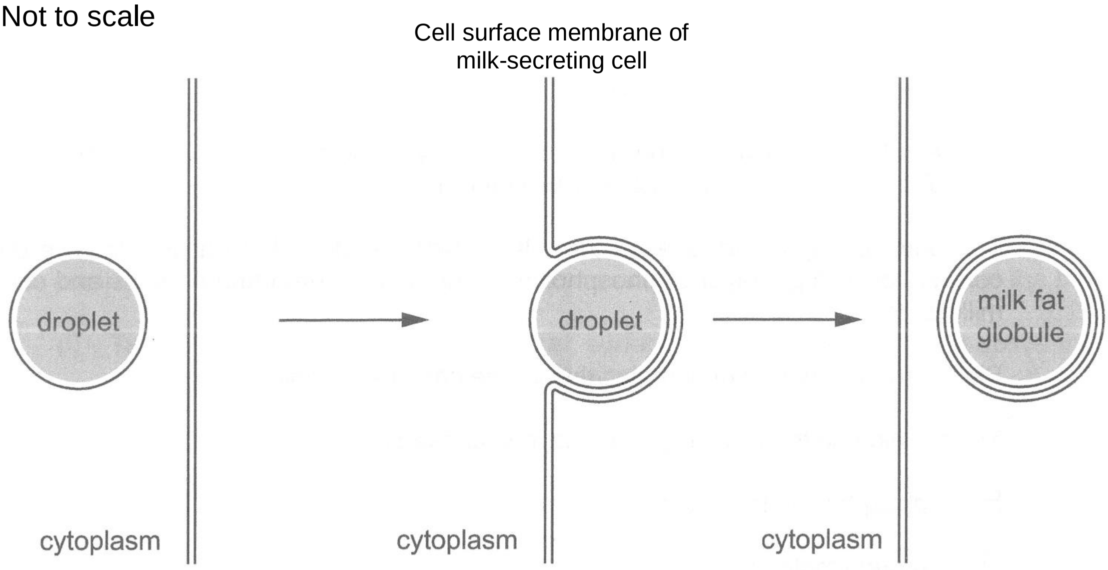


Fig. 1.3

The secreted milk fat globule has a unique membrane that has an outer phospholipid bilayer, then a layer of cytoplasm and inside that, a single phospholipid layer. Phospholipid bilayer is approximately 7nm thick while the layer of cytoplasm is approximately twice as thick as the phospholipid bilayer.

(ii) State the approximate thickness of the unique membrane of a milk fat globule.

....**[1]**

(iii) The stages in Fig. 1.3 are different from the exocytosis of secretory proteins.

Describe two differences between the stages in Fig. 1.3 and exocytosis of secretory proteins.

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.... **[2]**

(iv) Milk-secreting cells secrete lipids and glycoproteins at the same time. This helps the cells to maintain a constant cell surface membrane area.

Suggest why secreting lipids and glycoproteins at the same time helps to maintain a constant cell surface membrane area.

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(v) High temperature can damage cell membranes because of the denaturation of membrane proteins.

Describe how proteins become denatured at high temperature and explain how this could lead to damaging cell membranes.

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[Total: 25]

2. Burmese cats, *Felis catus*, show discontinuous variation in the colour of their eyes and hearing ability. A large scale investigation was conducted to study the inheritance pattern of eye colour and deafness in such cats.

Pure-breeding blue eyed cats which were deaf were crossed with pure-breeding yellow eyed cats with normal hearing.

The F_1 all had yellow eyes and normal hearing.

Female cats from the F_1 were crossed with male cats with blue eyes and were deaf. 668 offspring were produced.

The observed numbers of F_2 cats with each phenotype were as follows:

Blue eyes, deaf	259
Blue eyes, normal hearing	72
Yellow eyes, deaf	53
Yellow eyes, normal hearing	284

(a) Based on this study, state the expected results for F_2 cats from crossing F_1 female cats with male cats with blue eyes and deafness.

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A chi-squared test, with the guide of Table 2.1, was planned to evaluate the difference between observed and expected results.

Table 2.1

Degree of Freedom	**Probability, p**				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(b) With reference to Table 2.1, outline how the significance of the difference between observed and expected results can be evaluated with 98% confidence (Formulae not required).

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(c) The difference between observed and expected results was eventually concluded to be significant.

(i) Using the symbols A/a for eye colour and B/b for hearing ability, draw a genetic diagram to explain the observed results of the cross between F_1 female cats and male cats with blue eyes and deafness.

[4]

(ii) Explain why the observed phenotypic ratio is different from the expected phenotypic ratio.

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[Total: 10]

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3. Wheat is an important cereal crop in many parts of the world. Wheat seedlings were grown at three different concentrations of carbon dioxide (in parts per million, ppm) and the rate of photosynthesis was measured at various light intensities. Fig. 3.1 shows the results of the investigation.

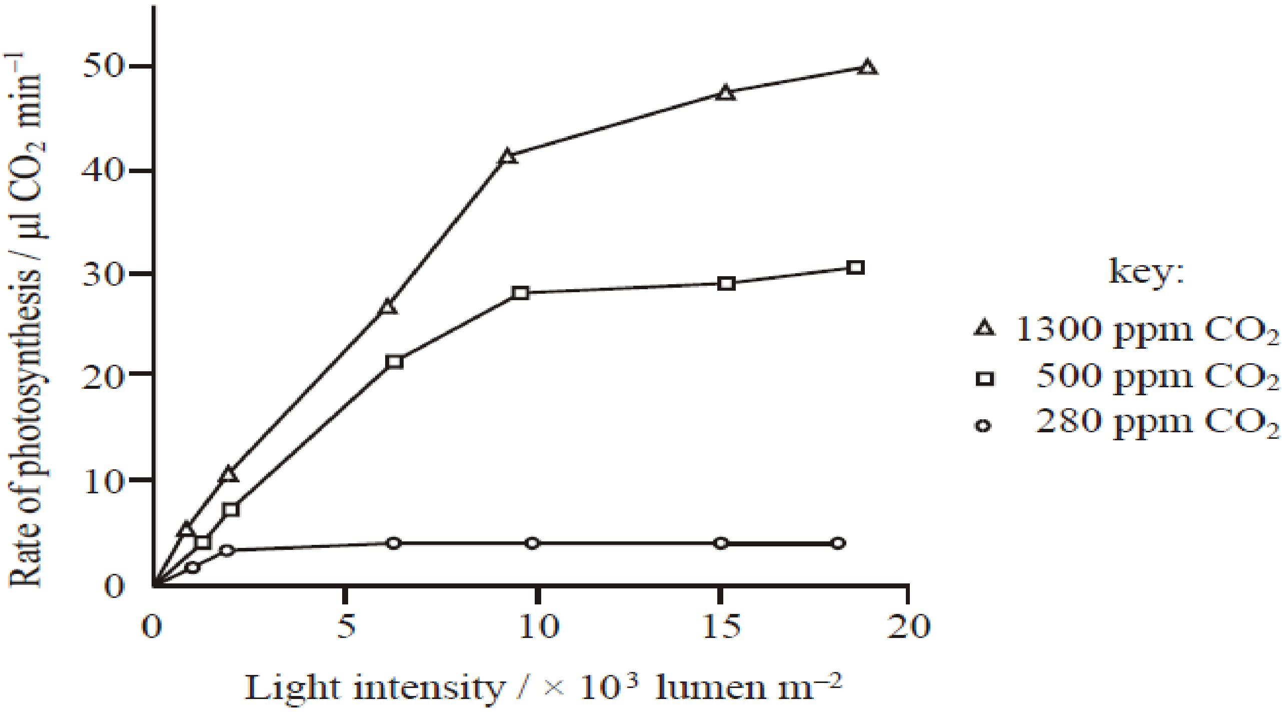

Fig. 3.1

(a) With reference to Fig. 3.1,

(i) describe the effect of light intensity and carbon dioxide concentration on the rate of photosynthesis from 5 x 10^3 lumen m^{-2} to 15 x 10^3 lumen m^{-2} for experiment conducted at 280 ppm and 500 ppm of CO_2.

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(ii) explain the trend observed at 1300 ppm CO_2.

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(b) ATP and NADPH are essential for the light-independent reactions of photosynthesis. Light-independent reactions of photosynthesis is also known as Calvin cycle.

(i) Describe the role of ATP and NADPH during Calvin cycle.

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(ii) One form of light-dependent reactions is the non-cyclic photophosphorylation. In some plant cells, cyclic photophosphorylation is more important.

Suggest how cyclic photophosphorylation resolves the limitation of non-cyclic photophosphorylation in Calvin cycle.

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Nicotinamide adenine dinucleotide (NAD) is a hydrogen acceptor used by many organisms. A novel compound is discovered to bind to NADH irreversibly in the cytosol.

(c) Suggest and explain the effect of this novel compound on aerobic and anaerobic respiration in mammals.

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[Total: 15]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous pose, where appropriate.

Your answers must be set out in parts (a) and (b), as indicated in the question.

4 **(a)** Explain why genetic variation may be lost during natural selection, and describe the ways of preserving genetic variation. **[15]**

(b) Describe the end-replication problem and discuss the importance of telomerase in specific human cell types. **[10]**

[Total: 25]

5 **(a)** Using specific examples, explain how phenotype can be influenced by both genotype and environment in prokaryotes and eukaryotes. **[15]**

(b) Explain how variation arises in viruses and discuss its impact. **[10]**

[Total: 25]

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